The new moment (50) identifies $\check{\lambda}$.[29] The advantage of the formulation in proposition 7 is that the variance V^θ is derived using standard GMM theory and implemented numerically using standard GMM routines. An analogous proposition can be stated for the OLS procedure, as in proposition 5, but adding the moment (50) to estimate $\check{\lambda}$. Then, unlike the positive message of proposition 5, when λ is estimated, the fact that z_t is a constructed regressor does affect the asymptotic variance of the estimator and we cannot rely on the OLS standard error. Instead, one must rely on the variance matrix V^θ, which is still easy to compute numerically.

C. *Heteroskedasticity*

We now discuss the case where the u_{it} are heteroskedastic. We call V^u their variance-covariance matrix. We assume for now that this is known, at least up to a factor of proportionality, and constant over time. Given a positive definite matrix W, define the $r \times N$ and $N \times N$ matrices,

$$R^{\lambda,W} := (\lambda' W \lambda)^{-1} \lambda' W, \qquad Q^{\lambda,W} := I - \lambda R^{\lambda,W}, \tag{52}$$

so that $Q^{\lambda,W}$ is a projection on the space orthogonal to λ and $R^{\lambda,W}$ is a projection on λ.[30]

Proposition 8 (GIV with heteroskedastic idiosyncratic shocks). Let us replace assumption 3 by assumption 4, assuming that we know V^u, and define the corresponding matrices $Q^{\lambda,W}$ and $R^{\lambda,W}$ in (52), with $W = (V^u)^{-1}$. Then propositions 1–7 and lemma 1 hold, provided that we change $\Gamma' S \neq 0$ into $\Gamma' V^u S \neq 0$ in (4); change Q^λ and R^λ into $Q^{\lambda,W}$ and $R^{\lambda,W}$ everywhere, for all occurrences of factors λ (e.g., in [17] we change $\Gamma^{*\prime} = S' Q$ into $\Gamma^{*\prime} = S' Q^{\lambda,W}$; in [20] we change R^λ, Q^λ into $R^{\lambda,W}$, $Q^{\lambda,W}$; in proposition 6, we change R^x into $R^{x,W}$); change equal weights E into quasi-equal weights $\tilde{E} := W\iota / \iota' W \iota$ (e.g., in [43]); and change (50) into $\Sigma_t(-\check{y}_t + \check{\lambda}\check{\eta}_t(m^y, \check{\lambda}) + \check{C}_t^y m^y) W \check{\eta}_t(m^y, \check{\lambda})' = 0$.

When idiosyncratic shocks are homoskedastic, $\tilde{E}_i = 1/N$, while if they are uncorrelated but heteroskedastic (i.e., $W = \mathrm{Diag}(1/\sigma_{u_i}^2)$), we have[31]

[29] Moment (50) means that the $r - 1$ column vectors $\check{\lambda}$ are eigenvectors of the empirical variance-covariance matrix of $\check{y}_t - \check{C}_t^y m^y$, so that a principal component analysis can be equivalently performed; see the proof of proposition 7. Instead of (51), one could use the moment $\Sigma_t(-\check{y}_t + \check{\lambda}\check{\eta}_t(m^y, \check{\lambda}) + \check{C}_t^y m^y)' \check{C}_t^y = 0$, but (51) simplifies the proof.

[30] These matrices have convenient properties that we record here (dropping the superscripts for simplicity):

$$Q\lambda = 0, \quad R\lambda = I, \quad Q' W \lambda = 0, \quad (I - Q) W^{-1} Q' = 0, \quad (I - Q') W Q = 0, \tag{53}$$
$$Q^2 = Q, \quad R W^{-1} Q' = 0.$$

[31] Generically, $\Gamma^* \neq 0$. But in cases where the variance is inversely proportional to size, $V^u S = a\iota$ for some scalar a, so that $\Gamma^{*\prime} = S' Q^{\lambda,W} = 0$ and the GIV would fail. This would be detected in practice via extremely large standard errors. Fortunately, in most contexts,

$$\tilde{E}_i := \frac{1/\sigma_{u_i}^2}{\sum_j 1/\sigma_{u_j}^2}. \tag{54}$$

In addition, in the cross-sectional regressions, we use generalized least squares with weights $W = (V^u)^{-1}$ instead of OLS (e.g., in [32]).

While estimating ψ could be done without knowing the $\sigma_{u_i}^2$, estimating ϕ^d, M, and μ does require the conditions (10) or (35) and hence some knowledge of the heteroskedasticity of the u_i's.[32]

Estimation of the degree of heteroskedasticity.—In the central case where the u_{it}'s are uncorrelated across i's, we can estimate the volatilities σ_{u_i} consistently, via the moments $\mathbb{E}[\check{u}_{it}^2] = Q_{ii}^{\lambda,W}\sigma_{u_i}^2$, where $\check{u}_t = Q^{\lambda,W}u_t$ is the vector of residuals; this fits in the GMM structure of the rest of the estimation.[33] This is detailed in appendix section D.3. When the σ_{u_i}'s are estimated, the standard errors of the estimators of ψ, ϕ^d, μ, and M adjust accordingly.

D. Generalization of the GIV to Other Setups

While we focus on the demand and supply setup for our main analysis, in this section we discuss how the basic ideas extend to a variety of settings.

Time-varying size weights.—Size weights could vary over time, $S_{i,t}$, so that in our leading example (2) the aggregate demand disturbance y_{St} becomes $y_{St} = \Sigma_i S_{i,t-1} y_{it}$. We then make the additional assumption that u_t is independent of S_{t-1}. Then, our identifying moments are still correct, replacing S_i by $S_{i,t-1}$ and Γ by Γ_t, with the natural changes (e.g., in [4] we replace S and Γ by S_t and Γ_t).[34]

Time-varying factors.—We can relax assumption 1 and replace $\lambda_i = X_i\dot{\lambda}$ by $\lambda_{it} = X_{it}\dot{\lambda}$. This would allow for time variation in the loadings on the aggregate shocks. In general, we can then have factor loadings that are observed and time varying and factor loadings that are unknown (see sec. IV.B). The combined loadings are then given by $\lambda_{it} = (X_{it}\dot{\lambda}, \lambda_i^F)$, where $X_{it}\dot{\lambda}$ fluctuates over time and λ_i^F is constant.

Multidimensional GIV.—The basic model can be extended to cover multidimensional outcomes y_{it} and shocks u_{it}. For instance, a firm i could have

variance may decay a bit with size S_i but not as fast as $1/S_i$ (see, e.g., Lee et al. 1998 and the discussion in Gabaix 2011).

[32] Recall that those conditions are identical when $\tilde{E} := S - \Gamma$. If we misspecify the variance of the u_{it}'s (keeping them uncorrelated), the impact on the estimates is typically quite small. For instance, if the $\sigma_{u_i}^2$'s are heteroskedastic and we ignore this fact, then $\mathbb{E}[u_{\Gamma t}u_{Et}] = (1/N)\Sigma_i\Gamma_i\sigma_{u_i}^2$ will be nonzero but will still be small when we have a large cross section, of order $O(1/N)$. The relative bias $\mathbb{E}[u_{\Gamma t}u_{Et}]/\mathbb{E}[u_{\Gamma t}^2] = O(1/Nh_N^2)$ more generally goes to zero when we are in a "granular" case where $h_N^2 \gg 1/N$ and where h_N is the Herfindahl (45)—for details, see proposition 2 in Gabaix (2011).

[33] The moment does feature $Q_{ii}^{\lambda,W}\sigma_{u_i}^2$, not $(Q_{ii}^{\lambda,W})^2\sigma_{u_i}^2$ as one might first think.

[34] The expressions of the variances will change—e.g., replacing σ_z^2 by its average value, which should remain bounded away from zero.

two-dimensional shocks u_{it}, one to productivity and one to labor demand. We show how GIVs can be used in this setting in appendix section D.6.

Beyond supply and demand for one good.—In appendix section D.5, we show how GIVs can be used to estimate parameters of interest in models that feature multiple general equilibrium channels.

Estimating structural vector autoregressions with GIVs.—We can estimate vector autoregressions and impulse responses with GIVs. If $Y_t = AY_{t-1} + X_t$ for vectors X_t, Y_t and matrix A, we can use the GIV z_t to instrument for some of the shocks to the innovations X_t and achieve partial or full identification. The GIV is then an "external instrument," and we can follow the methods spelled out in Stock and Watson (2018).[35] We can also estimate Jordà (2005)-style local projections, regressing $\mathbb{E}_t[Y_{t+h}] = \beta_h X_t$, and instrumenting some of the regressors X_t by GIVs. This shows how GIVs can be used to identify parameters in structural vector autoregressions, complementing an active literature that uses sign restrictions, as in Uhlig (2005), or narrative restrictions combined with sign restrictions, as in Antolín-Díaz and Rubio-Ramírez (2018) and Ludvigson, Ma, and Ng (2020).

Comparison with Bartik instruments.—Bartik (1991) instruments are widely used in economics, and we discuss the link between Bartik and GIV in appendix section D.4 and clarify the difference in identifying assumptions using the econometric framework of Borusyak, Hull, and Jaravel (2022).[36] We briefly summarize the main insights. First, in a number of cases where a cross section is studied, Bartik applies but GIV cannot be used as there is no large idiosyncratic shock that one can use (e.g., Autor, Dorn, and Hanson 2013). Alternatively, in a number of cases GIV applies naturally, particularly when there are large idiosyncratic shocks that affect aggregate outcomes. We therefore view Bartik and GIVs as complements in the toolkit of economists, and which empirical strategy is most appropriate depends on the setting.

When aggregate shocks are at least partially made of idiosyncratic shocks.—GIVs extend to economies where aggregate shocks η_t are themselves (at least partially) made of idiosyncratic shocks u_{it} (as in Long and Plosser 1983; Gabaix 2011; Acemoglu et al. 2012; Carvalho and Gabaix 2013; Carvalho and Grassi 2019; Brownlees and Mesters 2021). We develop this in appendix sections D.17 and D.18. These sections show that we can identify important parameters even if we have only crude proxies for the primitive shocks such as TFP.

[35] For recent developments in this area, see Plagborg-Møller and Wolf (2022) and Sarto (2022).

[36] Bartik instruments (also known as shift-share estimators) have been rigorously studied in recent econometric work; see also Adao, Kolesár, and Morales (2019); Goldsmith-Pinkham, Sorkin, and Swift (2020); Borusyak, Hull, and Jaravel (2022).

V. Conclusion

We developed granular instrumental variables (GIVs). The generative insight is that idiosyncratic shocks offer a rich source of instruments in concentrated economic environments with large idiosyncratic shocks. We lay out econometric procedures to extract them from panel data and efficiently aggregate them to obtain the most powerful instruments. We discuss various econometric extensions that might be useful in applied work. Given the ubiquity of concentration in various economic settings and the volatility of idiosyncratic shocks, we hope that GIVs provide a valuable addition to the toolkit of economists to estimate and understand causal relationships in the economy.

Appendix A

Proofs

This section shows some of the main proofs. Further proofs are in appendix C.

A1. Variance Facts

We will repeatedly use a number of facts that we record here. For two deterministic vectors X and Y of dimensions $n \times 1$, defining $u_X := X'u$ and $u_Y := Y'u$, we have

$$\mathbb{E}[u_X u_Y] = \mathbb{E}[(X'u)(Y'u)] = X'\,\mathbb{E}[uu']Y = X'V^u Y. \tag{55}$$

If $(u_i)_{i=1,\ldots,N}$ is a vector of uncorrelated random variables with mean zero and common variance σ_u^2, then $V^u = \sigma_u^2 I$ and

$$\mathbb{E}[u_X u_Y] = X'Y\sigma_u^2. \tag{56}$$

With $\Gamma = S - E$ (where $E_i = 1/N$), we have (with $h = \sqrt{\Sigma_{i=1}^N S_i^2 - (1/N)}$)[37]

$$\mathbb{E}[u_\Gamma u_E] = 0, \qquad \mathbb{E}[u_\Gamma^2] = \mathbb{E}[u_S u_\Gamma] = h^2\sigma_u^2. \tag{57}$$

In the general heteroskedastic case for V^u, defining $W = (V^u)^{-1}$, the quasi-equal weight vector is $\tilde{E} = W\iota/\iota' W\iota$. Then, for any Γ such that $\iota'\Gamma = 0$, we have[38]

$$\mathbb{E}[u_\Gamma u_{\tilde{E}}] = 0. \tag{58}$$

A2. Proof of Propositions 1 and 2

First, we note that (2) implies that

$$y_{St} = \phi^d p_t + u_{St} + \varepsilon_{1t},$$

[37] Indeed, $\Gamma'E = (1/N)\Sigma_i(S_i - E_i) = 0$, and $\Gamma'\Gamma = \Sigma_{i=1}^N[S_i - (1/N)]^2 = \Sigma_{i=1}^N[S_i^2 - (2/N)S_i + (1/N^2)] = \Sigma_{i=1}^N S_i^2 - (1/N) = h^2$.

[38] Indeed, as $\tilde{E} = kW\iota$ and $k = 1/\iota W\iota$, we have $\mathbb{E}[u_\Gamma u_{\tilde{E}}] = \tilde{E}'\,\mathbb{E}[uu']\Gamma = \tilde{E}'V^u\Gamma = k\iota'WV^u\Gamma = k\iota'\Gamma = 0$.